

Phase 3 — Model Development (Classification Systems)

Model A — Visit Risk Classification

Business Purpose: Predict whether a hospital visit represents a Low, Medium, or High operational and clinical risk.

- Define the target variable as `risk_score`.
- Select and justify feature set based on business relevance.
- Perform a time-based train and test split (earliest 80 percent for training, latest 20 percent for testing).
- Train a baseline model using Logistic Regression.
- Train an advanced model such as Random Forest or Gradient Boosting.
- Perform optional hyperparameter tuning and document results.

Model B — Claim Outcome Classification

Business Purpose: Predict whether an insurance claim will be Paid, Pending, or Rejected before submission.

- Define the target variable as `claim_status`.
- Engineer financial and operational predictor features.
- Perform a time-based train and test split.
- Train baseline and advanced classification models.
- Analyze class imbalance and mitigation strategy.

Deliverables

- Risk model notebook (`02_risk_model.ipynb`)
- Claim model notebook (`03_claim_model.ipynb`)
- Saved model artifacts (`.pkl` or `.joblib`)
- Feature schema file (`feature_schema.json`)